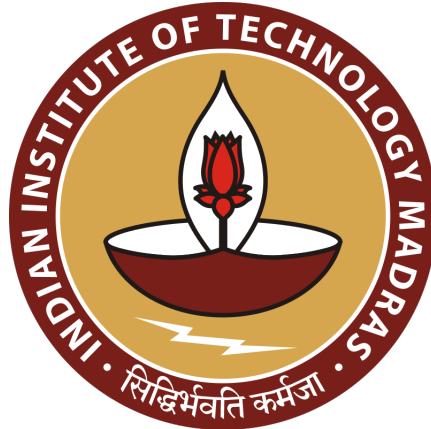


INDIAN INSTITUTE OF TECHNOLOGY, MADRAS



Data Science and AI Project

MILESTONE 2 SUBMISSION

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1 Dataset Collection and Preparation

1.1 Introduction

The objective of this project is to develop an Artificial Intelligence driven monitoring system for detecting and analyzing forest cover changes in the Aravalli region and the North-East region of India. Satellite imagery provides a reliable and scalable solution for monitoring environmental changes over time. In this project, multi-source satellite datasets are used to capture structural and spectral characteristics of vegetation.

The dataset used for this study is derived from three primary sources: Sentinel-1 Synthetic Aperture Radar (SAR) data, Sentinel-2 multispectral optical imagery, and the Hansen Global Forest Change dataset which provides labeled information about forest loss. These datasets are integrated to create a supervised learning dataset suitable for training deep learning models for forest change detection.

1.2 Dataset Sources

The dataset used in this project is constructed using the following publicly available satellite datasets.

1.2.1 *Sentinel-1 Dataset*

Sentinel-1 is a radar imaging satellite mission developed by the European Space Agency (ESA) under the Copernicus Earth Observation Programme. Sentinel-1 provides Synthetic Aperture Radar (SAR) imagery that can capture Earth's surface regardless of weather conditions or daylight availability. This makes it particularly useful for monitoring forest structures in regions with frequent cloud cover.

For this project, two polarization bands from Sentinel-1 are used:

- VV (Vertical transmit – Vertical receive)
- VH (Vertical transmit – Horizontal receive)

These radar backscatter bands provide information about surface roughness and vegetation structure. Forested regions typically produce strong VH signals due to the complex canopy structure, while deforested areas show reduced backscatter.

The official dataset documentation and access portal for Sentinel-1 can be found at:

- Copernicus Sentinel-1 Mission Overview: [Link](#)
- Sentinel-1 Dataset in Google Earth Engine: [Link](#)

1.2.2 *Sentinel-2 Dataset*

Sentinel-2 is a multispectral optical satellite mission also developed by the European Space Agency. It captures high-resolution imagery in multiple spectral bands, enabling detailed observation of vegetation and land cover characteristics.

The following spectral bands from Sentinel-2 are used in this project:

- B2 (Blue Band)
- B3 (Green Band)
- B4 (Red Band)
- B8 (Near Infrared Band)

These bands are widely used for vegetation monitoring. In particular, the Near Infrared band is highly reflective for healthy vegetation and is commonly used in vegetation indices such as NDVI.

The official dataset documentation and access links are provided below:

- Copernicus Sentinel-2 Mission Overview: [Link](#)
- Sentinel-2 Dataset in Google Earth Engine: [Link](#)

1.2.3 *Hansen Global Forest Change Dataset*

The Hansen Global Forest Change dataset is a widely used global dataset developed by researchers at the University of Maryland. It provides annual information on forest cover loss and gain at a spatial resolution of 30 meters.

For this project, the *lossyear* layer from the dataset is used to generate label masks for supervised learning. The *lossyear* variable indicates the year in which forest loss occurred at a specific pixel location.

The dataset documentation and access links are provided below:

- Global Forest Change Dataset Portal: [Link](#)
- Hansen Dataset in Google Earth Engine: [Link](#)

1.3 Dataset License and Usage Permissions

All datasets used in this project are publicly available and are distributed under open-access policies.

- Sentinel-1 and Sentinel-2 datasets are part of the Copernicus Programme and are freely available for research and commercial use.
- The Hansen Global Forest Change dataset is publicly released for scientific and educational purposes.
- Data processing and extraction are performed using Google Earth Engine, a cloud-based geospatial analysis platform that provides access to these datasets.

These datasets allow redistribution of derived datasets for research purposes, provided that proper attribution is maintained.

1.4 Dataset Structure

The dataset used for model training is constructed by combining Sentinel-1 and Sentinel-2 imagery along with the Hansen forest loss labels.

Each training sample consists of:

- Multi-channel satellite image tensor
- Corresponding forest loss label mask

The input image contains six channels corresponding to different satellite bands:

- **Sentinel-1 VV (Vertical Transmit – Vertical Receive)**

This radar band measures the strength of microwave signals that bounce back from the Earth's surface when transmitted and received in vertical polarization. It captures information about surface roughness and terrain structure. Areas with dense forest tend to produce stronger radar reflections compared to flat surfaces such as barren land or water bodies.

- **Sentinel-1 VH (Vertical Transmit – Horizontal Receive)**

This radar band is particularly sensitive to vegetation structure. When radar signals interact with complex forest canopies, they scatter in multiple directions, producing strong VH responses. Therefore, forested areas typically exhibit high VH backscatter, while cleared or deforested regions show lower values.

- **Sentinel-2 B2 (Blue Band)**

The blue band corresponds to wavelengths visible to the human eye. It is useful for identifying water

bodies, atmospheric haze, and surface reflectance properties. In the context of this project, it helps improve overall land cover discrimination.

- **Sentinel-2 B3 (Green Band)**

This band corresponds to green wavelengths and provides information about vegetation presence and general surface characteristics. Vegetation typically reflects moderate amounts of green light.

- **Sentinel-2 B4 (Red Band)**

The red band is strongly absorbed by healthy vegetation during photosynthesis. As a result, dense forests appear darker in this band. Areas with sparse vegetation or degraded forest tend to show higher reflectance values.

- **Sentinel-2 B8 (Near Infrared Band)**

The near infrared band is one of the most important bands for vegetation monitoring. Healthy vegetation reflects a large portion of near infrared radiation due to the internal structure of plant leaves. This band therefore provides a strong signal for identifying forest cover and distinguishing it from non-vegetated surfaces.

By combining these six channels, the dataset captures both structural information (from radar signals) and spectral information (from optical imagery), which together provide a comprehensive representation of forest conditions.

Each image sample has the following dimensions:

- Input Image: $6 \times 256 \times 256$
- Label Mask: $1 \times 256 \times 256$

The dataset is generated using bi-temporal satellite imagery from two different years (2022 and 2023). The model learns to detect forest loss by comparing temporal differences in satellite signals.

1.5 Classes and Categories

The dataset is designed as a binary classification problem at the pixel level.

- Class 0: No forest loss
- Class 1: Forest loss

Each pixel in the label mask indicates whether forest loss occurred during the specified year.

1.6 Dataset Size

Using the satellite data extraction pipeline implemented through Google Earth Engine, a total of **7500 training samples** were generated. Each sample corresponds to a spatial patch extracted from the region of interest.

Each 256×256 patch represents a small geographic region captured at approximately 10–30 meter spatial resolution. The label mask indicates whether forest loss occurred within the corresponding spatial region.

The patch-based approach significantly increases the number of training samples while maintaining manageable input sizes for deep learning models.

[Final Dataset on Kaggle](#)

2 Exploratory Data Analysis (EDA)

2.1 Sanity Checks

Before using the dataset for model training, an exploratory data analysis (EDA) stage was conducted to perform sanity checks on the generated satellite patches. Since satellite imagery can contain noise and environmental artifacts, it is important to ensure that the training dataset is clean and representative.

The purpose of the sanity check is to identify and remove corrupted or low-quality samples that could negatively impact model performance.

Several potential data quality issues were considered during this process.

- **High Cloud Coverage**

Optical satellite images from Sentinel-2 may contain clouds that obscure the land surface. Patches with excessive cloud coverage can hide vegetation signals and reduce the usefulness of the sample.

- **Cloud Shadows**

Even when clouds do not directly cover the region, cloud shadows may distort the spectral values of the underlying surface.

- **Water or Non-Forest Dominated Patches**

Some patches may contain large water bodies, barren land, or urban areas instead of forest regions.

These patches provide little useful information for forest change detection.

- **Sensor Noise or Missing Data**

Occasionally, satellite images may contain missing pixels, corrupted signals, or abnormal intensity values due to sensor errors or transmission issues.

- **Incorrect Label Alignment**

Misalignment between satellite imagery and the Hansen forest loss mask can lead to incorrect labels, which can negatively affect supervised learning.

- **Extreme Pixel Values**

Outliers in pixel intensity values may indicate processing errors or uncorrected atmospheric effects.

During the EDA process, statistical summaries and visual inspections were performed on randomly sampled patches to verify that the dataset contains valid vegetation signals and properly aligned labels.

Only after these sanity checks are completed will the dataset be finalized and used for training the deep learning model for forest cover change detection.

2.2 Satellite Image Visualization

Visual inspection of satellite imagery is an important step in understanding the characteristics of the dataset. Individual spectral bands from Sentinel-2 were visualized to observe vegetation patterns, land cover types, and possible areas of forest disturbance.

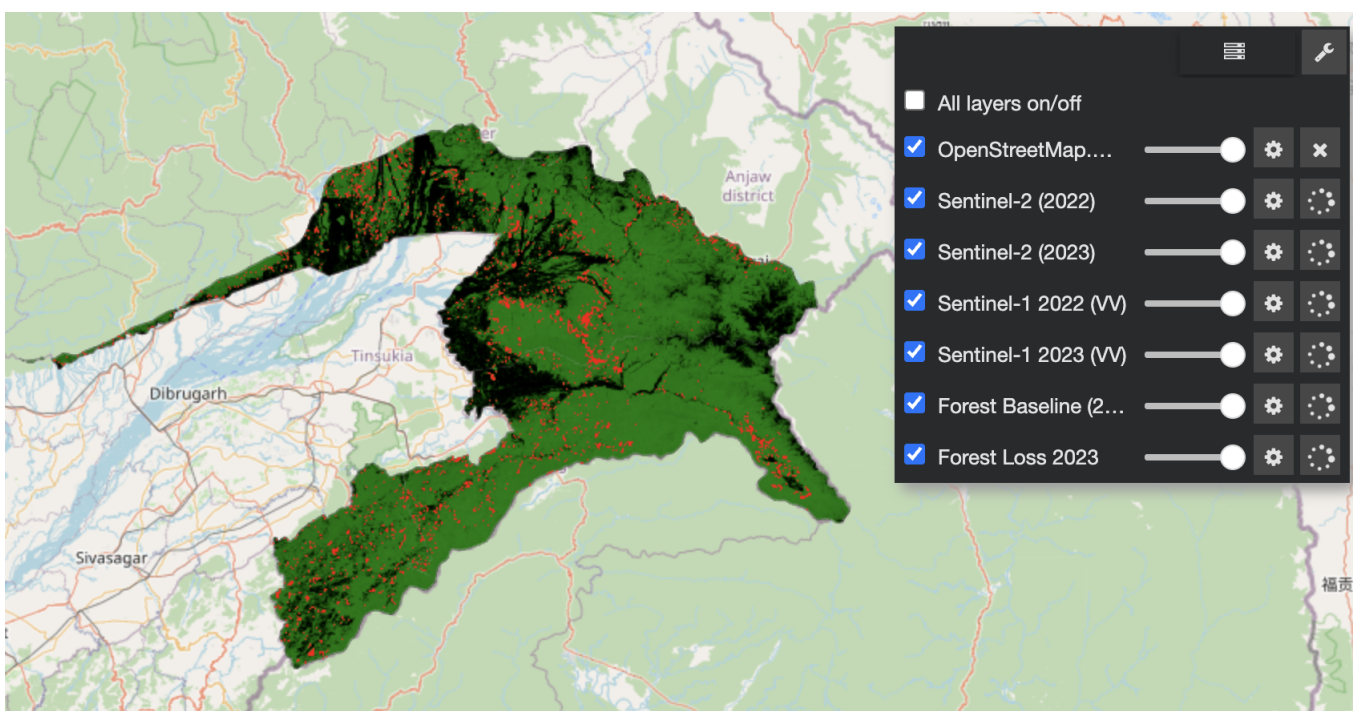


Figure 1: Satellite image visualization

Radar backscatter images from Sentinel-1 were also visualized to analyze structural characteristics of vegetation cover.

2.3 Forest Loss Mask Visualization

The forest loss mask obtained from the Hansen dataset was visualized to confirm the presence of deforestation signals in the selected region. Pixels representing forest loss were highlighted and compared with the satellite imagery.

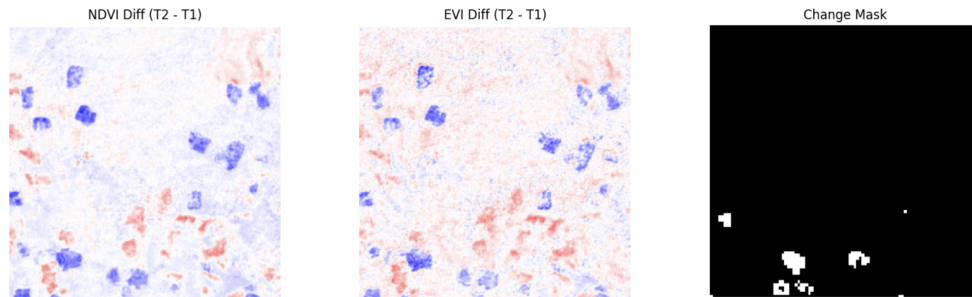


Figure 2: Temporal difference visualization

2.4 Band Distribution Analysis

Basic statistical analysis was performed on each spectral band to understand its distribution. Histograms and summary statistics were generated to analyze intensity variations across the dataset.

This analysis helps ensure that the data does not contain abnormal values or artifacts that may negatively affect model training.

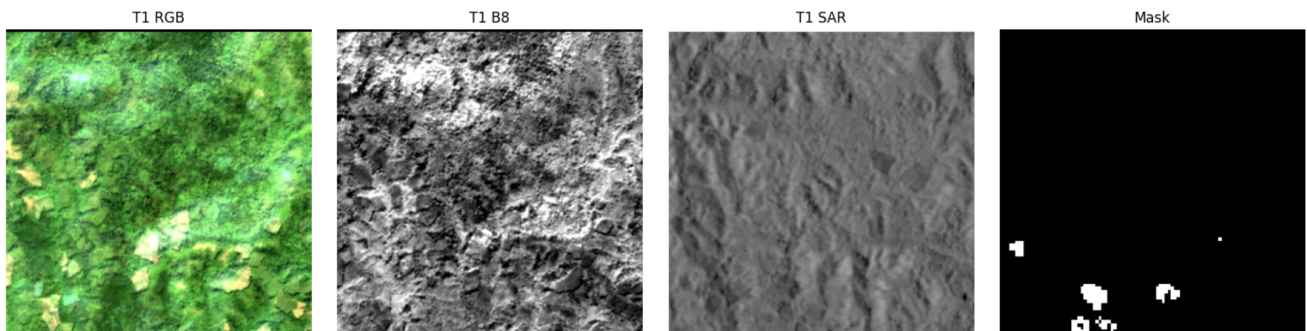


Figure 3: Band Visualization

3 Data Preprocessing

3.1 Data Extraction Using Google Earth Engine

Satellite imagery and forest loss masks were extracted using Google Earth Engine. The region of interest was defined using geographic coordinates, and images corresponding to the years 2022 and 2023 were collected.

Relevant spectral bands were selected and exported for further processing.

3.2 Patch Generation

Large satellite images were divided into smaller patches of size 256×256 pixels. This patch-based approach is commonly used in deep learning models for semantic segmentation because it allows efficient training and increases the number of training samples.

Each patch contains six spectral channels derived from Sentinel-1 and Sentinel-2 imagery.

3.3 Label Mask Creation

The label mask for each patch is generated from the Hansen forest loss dataset. Pixels where forest loss occurred in the year 2023 are labeled as class 1, while all other pixels are labeled as class 0.

This creates a binary segmentation problem where the model learns to identify areas of forest loss.

3.4 Data Normalization

To ensure consistent scaling across different spectral bands, pixel values were normalized before training. Normalization helps stabilize the training process and improves model convergence.

3.5 Final Dataset Preparation

The final dataset consists of multi-channel image tensors and corresponding label masks ready for training deep learning models. The dataset structure is organized in a format compatible with standard deep learning frameworks such as PyTorch and TensorFlow.

This dataset preparation pipeline ensures that the data is properly structured, labeled, and preprocessed for the subsequent modeling phase of the project.